Capturing Asymmetry and Heavy Tails in Bitcoin Price Volatility : A Comparative GARCH Family Analysis

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Abstract

Abstract. This paper examines Bitcoin's weekly return and volatility dynamics from March 2020 to March 2025 using a suite of GARCH-family models: the standard GARCH(1,1) and EGARCH(1,1) specifications, each under both Normal and Student's t distributions. Drawing on a comprehensive five-year dataset of weekly closing prices, we find clear evidence of volatility clustering, leverage effects, and fat-tailed return distributions. A model comparison based on Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and log-likelihood confirms that EGARCH(1,1) with a Student's t distribution provides the best overall fit. An 8-week out-of-sample volatility forecast highlights how distributional and leverage considerations significantly affect risk projections. These findings underscore the importance of asymmetry and heavy tails in accurately modeling Bitcoin's price behavior.

1 Introduction

Bitcoin has solidified its position as a leading cryptocurrency, with high levels of speculation and notoriously volatile price swings. Understanding and forecasting its volatility is vital for traders, investors, and risk managers. At the same time, Bitcoin's global appeal and decentralized nature make it particularly susceptible to market sentiment and external shocks, often resulting in dramatic price movements. Moreover, rapid innovation in the broader crypto space has introduced new competing digital assets and derivatives, amplifying Bitcoin's role as a benchmark for cryptocurrency volatility. Although GARCH-type models are well-suited to capturing time-varying volatility and clustering in financial markets, the extent to which different distributions (Normal vs. Student's t) and formulations (GARCH vs. EGARCH) improve model fit and forecast accuracy in Bitcoin remains an open question.

In this paper, we undertake an econometric analysis of Bitcoin's weekly returns over a five-year horizon (March 2020–March 2025), using:

1. GARCH(1,1) with Normal errors,
2. GARCH(1,1) with Student's t errors,
3. EGARCH(1,1) with Normal errors,
4. EGARCH(1,1) with Student's t errors.

After comparing the in-sample fit with AIC, BIC, and log-likelihood, we provide an 8-week out-of-sample volatility forecast. Our results offer deeper insight into how asymmetry and heavy tails should be considered when modeling a highly speculative asset such as Bitcoin.

2 Methodology

2.1 Data Collection

This study uses Bitcoin weekly closing price data from March 2020 to March 2025, sourced from reputable cryptocurrency market data providers. The initial dataset is checked for missing values, irregularities, and duplicated time entries. We compute both simple and logarithmic returns:

$$\text{Weekly Return} = \frac{P_t - P_{t-1}}{P_{t-1}}, \qquad \text{Log Return} = \ln\left(\frac{P_t}{P_{t-1}}\right). \tag{1}$$

2.2 Preliminary Analysis and Visualizations

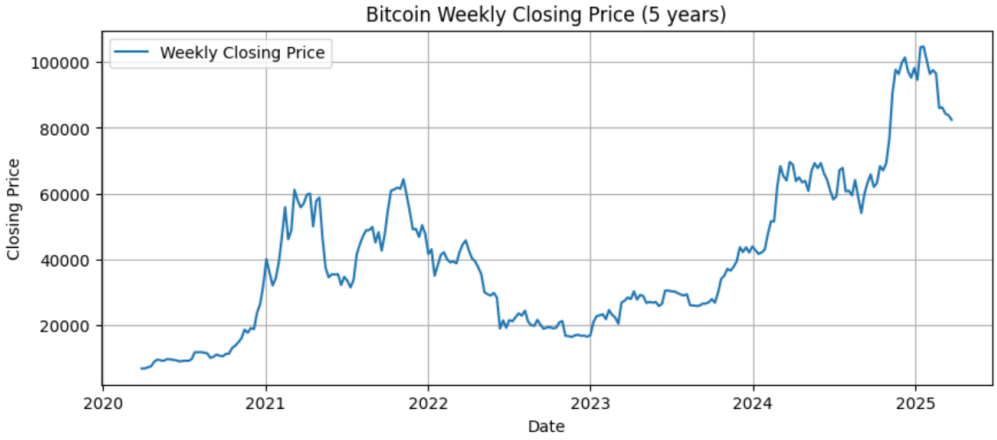


Figure 1: Weekly Closing Price Plot for 5 years (March 2020 – March 2025)

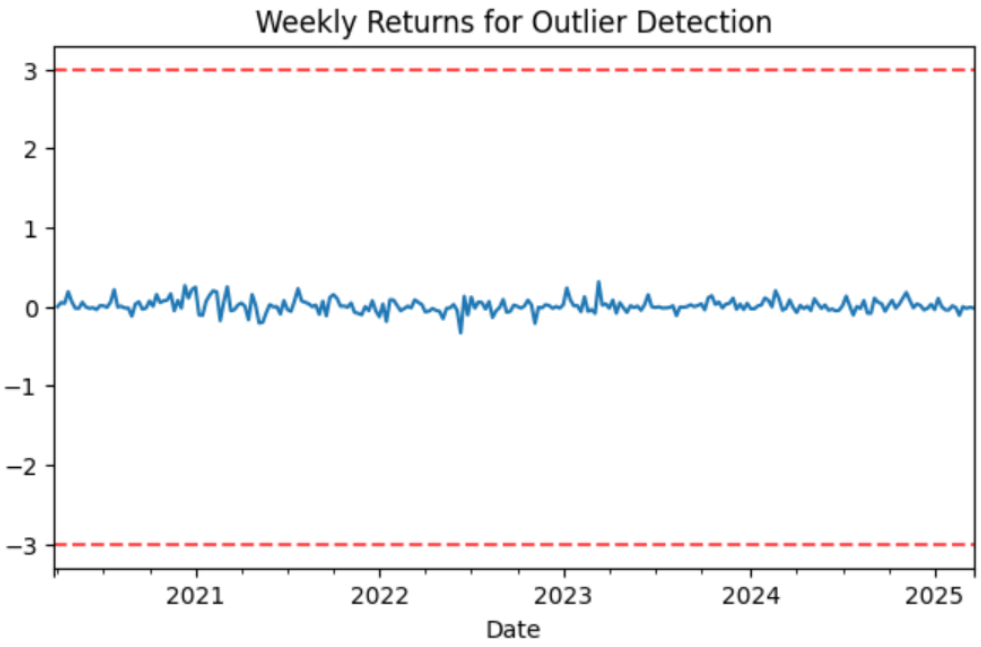


Figure 2: Weekly Returns with $\pm 3\sigma$ Limit Checks

Figure 1 illustrates the overall upward trajectory punctuated by sharp price swings. Figure 2 shows weekly returns and highlights potential outliers exceeding $\pm 3\sigma$. Such extreme events underscore the possibility of heavy-tailed behavior in Bitcoin returns.

Date	Close	Weekly_Return	Log_Return	Weekly_Volatility
2020-03-29	6857.4	NaN	NaN	NaN
2020-04-05	6867.8	0.001517	0.001515	NaN
2020-04-12	7230.8	0.052855	0.051506	NaN
2020-04-19	7540.4	0.042817	0.041926	NaN
2020-04-26	8966.3	0.189101	0.173198	0.074016
2020-05-03	9554.6	0.065612	0.063550	0.061080
2020-05-10	9379.5	-0.018326	-0.018496	0.080027
2020-05-17	9177.0	-0.021590	-0.021826	0.091668
2020-05-24	9692.5	0.056173	0.054652	0.045926
2020-05-31	9669.6	-0.002363	-0.002365	0.035474

Figure 3: Selected Feature Transformations (Returns, Log Returns, Rolling Volatility)

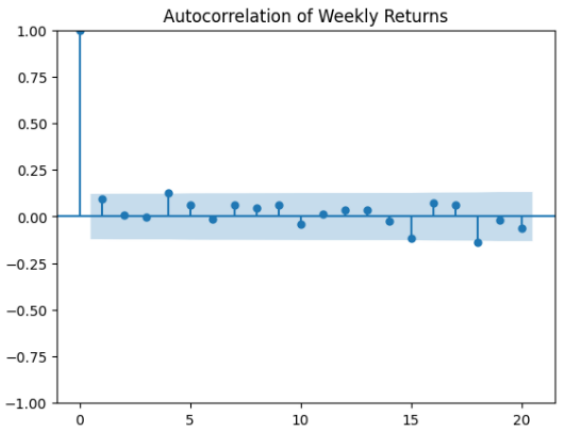
	Weekly_Return	Weekly_Volatility
count	260.000000	257.000000
mean	0.013202	0.071297
std	0.085379	0.042803
min	-0.331541	0.007506
25%	-0.030204	0.041977
50%	0.005125	0.062489
75%	0.055955	0.091693
max	0.314968	0.247762

Figure 4: Example Statistical Summary

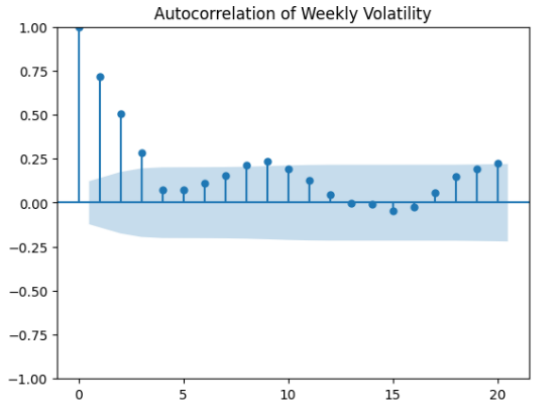
Descriptive statistics (mean, standard deviation, skewness, kurtosis) further confirm that the return series is not normally distributed, showing high kurtosis (fat tails).

2.3 ACF Analysis: Returns vs. Volatility

We examine autocorrelation functions (ACF) to gauge linear dependence in weekly returns and volatility (often proxied by squared returns). While the ACF of returns displays no significant autocorrelation beyond the immediate lag, the volatility measure exhibits persistent, slowly decaying autocorrelation. This persistence implies that once the market enters a high-volatility state, it tends to remain volatile for consecutive weeks. Conversely, low-volatility regimes may also endure until a new shock arrives. Such behavior is characteristic of financial markets subject to investor herding and rapid sentiment shifts, and it strongly motivates the use of GARCH-family models that explicitly account for time-varying conditional variance. Figure 5 shows ACF plots for returns (Panel A) and volatility (Panel B).



(a) ACF of Weekly Returns



(b) ACF of Weekly Volatility (Squared Returns)

Figure 5: Autocorrelation Analysis

These findings motivate the use of GARCH-family models, which model time-varying volatility and capture clustering effects.

3 Econometric Modeling

We consider four distinct GARCH-family models:

1. **GARCH(1,1) with Normal errors**:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad \epsilon_t \sim N(0, \sigma_t^2).$$

2. **GARCH(1,1) with Student's t errors**:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad \epsilon_t \sim t_\nu(0, \sigma_t^2).$$

3. **EGARCH(1,1) with Normal errors**:

$$\ln(\sigma_t^2) = \omega + \alpha\left|\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right| + \gamma\left(\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right) + \beta\ln(\sigma_{t-1}^2).$$

4. **EGARCH(1,1) with Student's t errors**:

$$\ln(\sigma_t^2) = \omega + \alpha\left|\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right| + \gamma\left(\frac{\epsilon_{t-1}}{\sigma_{t-1}}\right) + \beta\ln(\sigma_{t-1}^2), \quad \epsilon_t \sim t_\nu.$$

EGARCH allows for asymmetry (via γ) and does not require σ_t^2 to be strictly positive, since it is modeled in log form. Student's t helps capture heavier tails compared to a Normal distribution.

4 Empirical Results

4.1 Model Comparison: AIC, BIC, and Log-Likelihood

Table 1 reports the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and maximized log-likelihood (LogLik) for each specification. Lower (more negative) AIC/BIC and higher log-likelihood generally indicate superior fit.

Table 1: Comparison of Model Criteria

Model	**AIC**	**BIC**	**LogLik**
GARCH(1,1) Normal	−549.31	−535.07	278.66
GARCH(1,1) Student's t	−575.82	−558.02	292.91
EGARCH(1,1) Normal	−553.49	−535.69	281.75
EGARCH(1,1) Student's t	−578.92	−557.56	295.46

Note: Bold values indicate the best fit among the four models.

These criteria suggest that **EGARCH(1,1) with Student's t** offers the strongest overall fit, indicating the importance of allowing both asymmetric volatility and heavy tails for Bitcoin data.

4.2 Volatility Forecasting: 8-Week Horizon

An 8-week out-of-sample forecast provides insight into each model's short-term volatility expectations. Table 2 shows the predicted standard deviations (weekly) for horizons $h = 1$ to $h = 8$. We use an `adjustbox` environment to ensure full visibility.

Table 2: 8-Week Volatility Forecast by Model (Weekly)

Model	**Forecast Horizon (Weeks)**							
	h=1	**h=2**	**h=3**	**h=4**	**h=5**	**h=6**	**h=7**	**h=8**
GARCH(1,1) Normal	0.0676	0.0679	0.0682	0.0685	0.0688	0.0690	0.0693	0.0694
GARCH(1,1) Student's *t*	0.0736	0.0747	0.0756	0.0768	0.0780	0.0792	0.0801	0.0806
EGARCH(1,1) Normal	0.0645	0.0647	0.0648	0.0649	0.0650	0.0651	0.0652	0.0653
EGARCH(1,1) Student's *t*	0.0899	0.0899	0.0902	0.0901	0.0903	0.0904	0.0906	0.0908

Note: Forecasted weekly standard deviations (decimal form). For annualized volatility, multiply by $\sqrt{52}$.

Notably, EGARCH(1,1) Student's t tends to predict higher near-term volatility, reflecting its sensitivity to possible large shocks. Normal-based models produce comparatively milder forecasts.

5 Discussion

Our results underscore two key aspects of Bitcoin's market behavior:

1. **Heavy-tailed returns:** Student's t outperforms Normal in capturing the extremes often observed in cryptocurrency data.
2. **Asymmetric (Leverage) Effects:** EGARCH's additional parameter γ permits negative shocks to have disproportionate effects on volatility, aligning with the sentiment-driven nature of Bitcoin trading.

From a risk management perspective, relying solely on a Normal-based GARCH specification may understate the probability of tail events. The more conservative EGARCH(1,1) Student's t forecasts could be valuable for short-term hedging and stress testing. This aligns with existing literature suggesting that crypto markets are prone to abrupt, high-magnitude price movements.

6 Conclusion

Using weekly data from March 2020 to March 2025, this paper compares four GARCH-family models—spanning Normal vs. Student's t distributions and GARCH vs. EGARCH specifications—to analyze Bitcoin's volatile price dynamics. The empirical evidence shows that **EGARCH(1,1) with Student's *t*** best captures asymmetry and heavy tails, leading to superior in-sample fit (lowest AIC/BIC, highest log-likelihood) and more conservative short-term volatility forecasts.

Future Research

Further studies could incorporate external factors (e.g., macroeconomic indicators, regulatory announcements, on-chain analytics) into multivariate frameworks, or compare these econometric models to machine-learning approaches for enhanced predictive power over longer horizons.

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